Remote sensing image registration based on fusion of spatial transformation and dense convolution

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***ABSTRACT*-In order to achieve high-precision registration of remote sensing images, this paper proposes a remote sensing image registration method based on spatial transformation and dense convolution fusion. First, the dense convolutional network improved by the spatial transformation module is used for feature extraction to improve the robustness of feature points; then use improved Grid-based Motion Statistics (GMS) algorithm for feature matching, and use the homography matrix-based method to eliminate mismatched point pairs to improve matching accuracy; finally, the transformed model is used to solve the transformed model to achieve accurate registration of remote sensing images. Experiments show that the proposed method can effectively improve the correct matching rate of feature points, has higher registration accuracy and stronger robustness.**

Keywords-Dense neural networks; Improved Grid-based Motion Statistics algorithm; Remote sensing image registration

I. INTRODUCTION

Remote sensing image registration technology has been widely used in aerial image mapping, military situation perception and other fields. Differences in imaging conditions and complex ground features have exacerbated the geometric deformation between the images, which has made the registration of remote sensing images more difficult. When using a gray-based method to register such images, the area where the scene changes in the image will cause large gray-scale fluctuations, resulting in poor algorithm robustness. Feature-based registration method performs registration by calculating the correspondence among salient points, lines, contours and other features between images. It has strong robustness and is an important technical method in remote sensing image registration.

Among feature-based registration methods, many scholars have studied the registration method based on SIFT (scale invariant feature transform) feature points. Sedaghat et al. [1] proposed an adaptive double-scale invariant feature transform (AB-SIFT) algorithm for automatic matching of remote sensing images under local deformation. Xiang et al. [2] used an improved SIFT algorithm (OS-SIFT) to achieve image registration from optical to synthetic aperture radar (SAR), which has a good registration effect on optical SAR images. However, in the remote sensing image, due to severe appearance changes and abnormal areas, the effect of feature point detection and matching may be affected. At the same time, SIFT-based methods need to be improved in extracting spatial geometric information, color information, and high-level semantic information, which limits Application of image registration.

In recent years, deep learning and neural networks (neural network) have been widely used in image classification, image segmentation, and image registration. Quan and Wang et al. [3] designed a deep neural network for SAR image registration. This method has good registration performance in SAR image registration experiments. Zampieri et al. [4] proposed a non-rigid image registration method from coarse to fine using neural networks. Zhang et al. [5] used Amulet, a multi-layer convolution feature framework, for salient target detection. Li et al. [6] performed convolution feature fusion through an improved multi-scale Fisher kernel coding method. Huang et al. [7] realized the feature reuse and information transfer by building a dense convolutional network model, which improved the effect of neural network feature extraction. These studies show that the use of deep neural networks to effectively improve the registration effect of remote sensing images has become an effective technical means in the field of registration.

In order to enhance the ability of neural networks to extract features from remote sensing images and improve the robustness of the network, a new network model STN-DenseNet is constructed for feature extraction. This model combines a dense convolutional network with a spatial transformation network (STN). On the basis of DenseNet-121, by adding a spatial transformation module after the input layer, the input image or learned features can be spatially aligned, which improves the spatial invariance of the model. Using this model for remote sensing image feature extraction can effectively enhance the geometric invariance of convolution features and improve the robustness of feature points. The improved Grid-based Motion Statistics (GMS) algorithm is then used to match the features extracted by the STN-DenseNet model, and the method based on the homography matrix is used to eliminate mismatched point pairs to improve the matching accuracy. Finally, the transformed model is solved by using the correct matching point pairs after screening, so as to achieve accurate registration of remote sensing images.

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II. IMPROVE IMAGE REGISTRATION FOR DENSE CONVOLUTIONAL NERWORKS

A. Improved STN-DenseNet Model

The spatial transformation network (STN) [8] can effectively improve the translation invariance of the convolutional neural network, at the same time, it can also locate the target area of the image, and transform the area to the optimal state, enhancing the registration effect of the deformed area of the remote sensing image. DenseNet uses the Dense Block module to effectively improve the training effect and avoid overfitting. Therefore, in order to enhance the neural network's feature extraction capability for remote sensing images, a new network model STN-DenseNet is constructed for feature extraction. This model combines a dense convolutional network with a spatial transformation network (STN), on the basis of DenseNet-121, adds an STN module after the input layer, and aligns the input image or learned features in space to enhance the geometric inconsistency of the convolution features Degeneration and robustness. The model set up in this paper includes a STN module, 4 Dense Blocks, 4 convolution layers connected to Dense Blocks, and 4 pooling layers. Except for 1×1 convolutions in dense convolution blocks, the convolution kernel size in the other convolution layers is set to 3×3. The first three pooling operations use 2×2 average pooling, and the last pooling operation uses 2×2 global average pooling.

Traditional convolutional neural networks maintain the invariance of the network through convolution and pooling operations, but in fact, the characteristics are not unchanged after convolution and pooling operations. STN-DenseNet adds the STN spatial transformation module after the input layer. This part The structure based on the heuristic affine transformation matrix can be used to improve the overall invariance of the network, and the image can be cropped, translated, rotate, stretch, and distort the pixels that form the output feature image to improve the ability of the neural network to read the data.

Assuming that the input image is U and the output feature image is V, the formulas for the STN module to modify the image are shown in (1) and (2):

$$\begin{pmatrix} x_i^s \\ y_i^s \end{pmatrix} = T_\theta(G_i) = A_\theta \begin{pmatrix} x_i^t \\ y_i^t \\ 1 \end{pmatrix} = \begin{bmatrix} \theta_{11} & \theta_{12} & \theta_{13} \\ \theta_{21} & \theta_{22} & \theta_{23} \end{bmatrix} \begin{pmatrix} x_i^t \\ y_i^t \\ 1 \end{pmatrix} \tag{1}$$

$$V_i^c = \sum_n^H \sum_m^W U_{nm}^c \max\left(0,1-\left|x_i^s - m\right|\right) \max\left(0,1-\left|y_i^s - n\right|\right) \tag{2}$$

Among them, $T_\theta(G_i)$ is the feature map, A_θ is the affine transformation matrix, $\left(x_i^s, y_i^s\right)$ is the coordinates of each pixel of the input feature image U, and $\left(x_i^t, y_i^t\right)$ is the coordinates of each pixel of the output feature map V. H and W respectively represent the height and width of the feature, V_i^c is the output coordinate sequence, and U_{nm}^c is the coordinate sequence in the input color channel.

At the same time, DenseNet's feature transfer method is different from other convolutional neural networks, allowing DenseNet to be better trained after merging with STN networks, and has obvious enhancement effects. Because each layer of DenseNet and its Dense Block are connected to the previous layer, the features extracted by each layer of the network are equivalent to a non-linear transformation of the input data to directly obtain the input features. The layer-by-layer transmission method, the network layer behind the DenseNet network can get more original features, effectively prevent overfitting and greatly solve the gradient dispersion, and can better train the STN part of the model to achieve end-to-end Joint training. STN-DenseNet not only solves the problem of affecting training results due to data sensitivity, but also can extract features better through its spatial conversion module.

B. Loss Function

The method of model training reference [9] uses contrastive loss L to train the model. This loss function can effectively deal with the relationship of image block pairs in the model. The expressions of continuous loss are shown in (3) and (4):

$$L = \frac{1}{2N}\sum_{n=1}^{N} YD_W^2 + (1-Y)\max\left(m - D_W, 0\right)^2 \tag{3}$$

$$D_W(X_1, X_2) = \left(\sum_{i=1}^{P}\left(X_1^i - X_2^i\right)^2\right)^{\frac{1}{2}} \tag{4}$$

Among them, D_W represents the Euclidean distance between two sample features X_1 and X_2, P represents the feature dimension of the sample, Y is a label whether the two samples match, $Y=1$ represents the two samples are similar or match, $Y=0$ represents a mismatch, m is the set threshold, and N is the number of samples.

The model training process sets 100 iteration cycles, the model growth rate is set to 12, and the learning rate is 0.0009. In order to make the model converge quickly, the stochastic gradient descent (SGD) algorithm is used in the reverse transfer stage.

C. Improved GMS Algorithm for Feature Matching

After using STN-DenseNet model for feature extraction, this paper uses an improved Grid-based Motion Statistics (GMS) algorithm to perform initial matching on the feature points of the reference image and the image to be registered. The GMS algorithm [10] is an application of consistency constraints. It uses smooth constraints to integrate multiple matching information to make high-quality decisions to eliminate mismatches and improve the accuracy of matching. It is a highly robust image. Matching algorithm.

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This article meshes the input image to generate $G = O \times Q$ unit grids, where O is the number of vertical grids and Q is the number of horizontal grids. In order to reduce the amount of calculations and improve the robustness of the algorithm, for each element grid, an improvement is made based on the nine-grid statistical method of the GMS algorithm. Only the feature scores of the current grid and the adjacent and symmetrical 4 corner grids are counted. The sum of the feature scores of the five grids is called the cross-grid feature score S_i ,as shown in formula (5):

$$S_i = \sum_{j=1}^{5} s_{i,j} \tag{5}$$

Among them, $s_{i,j}$ is the feature score of the i grid in the forked grid where the j grid is located.

In order to improve the matching effect of the algorithm on the rotation area, this paper rotates the fork grid clockwise. It only needs to count the maximum mesh feature score S_i^R in the case of 4 rotations. Compared with the original GMS algorithm that requires 7 rotations, it reduces the calculation by more than half. S_i^R calculation formula is shown in (6):

$$S_i^R = \max\left\{\sum_{j=1}^{5} S_{i,j}^k, k = 0, \cdots, 3\right\} \tag{6}$$

Among them, S_i^R is the feature score of the k grid in the fork grid where the i grid is located after j the rotation.

Count the mean of the number of coarse matching features of the grid in the current fork lattice W_i , and filter out the correct matching point pairs according to the grid feature score threshold S_T . The calculation formulas are shown in (7) and (8):

$$W_i = \frac{1}{5}\sum_{j=1}^{5} W_{i,j} \tag{7}$$

$$S_T = \alpha\sqrt{W_i} \tag{8}$$

Among them, in the formula, $W_{i,j}$ is the number of coarse matching features in the i grid in the cross grid where the j grid is located, and α is the weight sparse coefficient, which is generally set to 6. When the maximum grid feature score S_i^R is greater than the grid feature score threshold S_T , it is the correct matching point selected by the GMS algorithm. Use this step to filter out all the correct matching points in the image.

In order to further improve the matching accuracy, this paper uses a method based on the homography matrix to eliminate mismatched points. Randomly extract 4 pairs of matching point pairs obtained previously, calculate the homography matrix F between the reference image and the image to be registered, and use it as the reference matrix. The remaining matching point pairs are calculated according to the distance function d , and the point pairs smaller than the threshold δ are regarded as the final correct matching points. The calculation formula is shown in (9):

$$d\left(x, x^{'}\right) = \left\|x - Fx^{'}\right\| + \left\|x^{'} - F^{-1}x\right\| \tag{9}$$

Among them, x and $x^{'}$ are corresponding points in the reference image and the image to be registered, $\|*\|$ is the Euclidean distance between two points, for a pair of matching points A_i and B_i , if $d\left(A_i, B_i\right) > \delta$, A_i and B_i are considered to be a correct pair of matching points; otherwise, the pair of matching points is eliminated. The δ value is set to 1.8, which is based on experimental experience.

The least squares method is used to calculate the transformation model based on the filtered correct matching points, and the images to be registered are corrected to complete the accurate registration of the remote sensing images.

D. Specific Steps of This Method

The specific research ideas of the remote sensing image registration method of STN improved dense convolutional network mentioned in this article are as follows:

Step 1 Image preprocessing. It mainly includes input image data and model training data processing.

Step 2 Feature extraction. Using the trained STN-DenseNet model to extract features from the input image can effectively enhance the geometric invariance of convolution features and improve the robustness of feature points.

Step 3 Build the feature descriptor. The method of reference [11] uses Kronecker product to construct the feature descriptors of the Dense Block 2, 3, and 4 layers in the STN-DenseNet model.

Step 4 Feature matching. After using the STN-DenseNet model for feature extraction, this paper uses the GMS algorithm to perform rough matching on the feature points of the reference image and the image to be registered, and initially selects valid matching points.

Step 5 Exact match. The method based on homography matrix is used to eliminate mismatch points and improve the matching accuracy of feature points.

Step 6 Solve the transformation model. Use the final correct matching points to calculate the image transformation matrix and transform the images to be registered to complete the accurate registration of the remote sensing images.

III. EXPERIMENTAL RESULTS AND ANALYSIS

The algorithm in this article is compiled on Spyder using python code. The compilation environment is TensorFlow 1.8.0 under Windows 10, and the CUDA and cudnn versions

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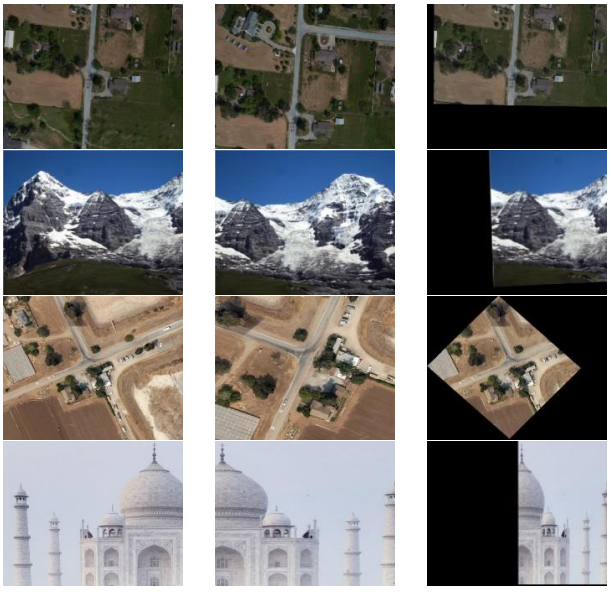
are 9.2.0 and 7.4.2 respectively. Experimental hardware environment: Intel Core I7-7700K CPU; Nvidia GTX1080Ti GPU; 16G dual DDR4 memory.

A. Data Set

The dataset was constructed using 59 sets of remote sensing image pairs obtained from Google Earth. According to the image category, it can be divided into the following categories: field remote sensing images, urban remote sensing images, low-altitude drone images, and landscape images. The largest proportion is the first category, and the remaining three categories have a small proportion and the same number. The remote sensing images in the field are mainly remote sensing images of lakes and rivers with clear boundaries. On the 59 pairs of images, 2096 matching "seed" image blocks were selected, each of which has a size of 32×32 pixels. Each of these "seed" tile pairs is declared as representing its own class. To extend these classes, 210 random transforms are applied to each "seed" image block pair. Each transformation is a combination of multiple random basic transformations, including rotation, translation, scaling, and brightness. Therefore, the custom data set constructed in this article has a total of Z classes, each class contains 2K samples, and is randomly divided into a training set and a test data set according to a 4: 1 ratio.

B. Image Registration Experiment Analysis

The experiment used four sets of remote sensing images obtained from Google Earth. Figure.1 shows the registration results of the algorithm for four groups of remote sensing images. The size of each group of images is 640×427, 640×427; 400 ×300, 400 ×300; 480 ×411, 480 ×439; 708 ×633, 548 ×628. (Unit: pixels)



(a)reference images (b) sensed images (c) results

Fig.1 Remote Sensing images

Use the STN-DenseNet model to extract these four sets of image features and obtain 3,000 rough matching point pairs through brute force matching. The traditional GMS algorithm, random sampling consistency algorithm (RANSAC) and the improved GMS algorithm in this paper are used to compare the accuracy of feature point matching. Table 1 shows the average matching accuracy rate and feature matching operation time (unit: milliseconds) of the four groups of images.

Table 1. Matching correct rate (%) and running time(ms)

Method	Match accuracy	Running time
RANSAC	96.38	118.47
GMS	97.24	25.92
OURS	97.58	18.4

From the experimental results in Table 1, it can be seen that after the RANSAC algorithm eliminates feature point mismatches, the matching accuracy rate (the number of correct matching points / the number of all matching points) is the lowest, and due to the complexity of its own algorithm, the operation time is the longest and the average time is consumed. 118.47ms. After the GMS algorithm eliminates false matches, the matching accuracy rate is higher than the RANSAC algorithm, and the average time is 25.92ms. The algorithm in this paper improves the GMS algorithm, which reduces the calculation time and increases the number of correct matching points. The average matching accuracy rate was the highest at 97.58%. The average time is 18.41ms, which is 29.01% lower than the GMS algorithm.

The literature [12] and [13] and the algorithm in this paper are used to compare the image registration accuracy. Verify the effectiveness of the algorithm in this paper. Figure.2 shows the stitching results after registration of the three algorithms. Table 2 shows the registration accuracy results of the three algorithms.

Figures 2 (a) ~ 2 (c) are the final stitching results of the four sets of images in the literature [12] and the literature [13] algorithm and the algorithm in this paper. Among the parts selected by the yellow boxes, the stitching effect can be clearly seen. Literature [12] uses SIFT algorithm to extract feature points, and uses RANSAC algorithm to perform precise matching, which effectively solves the problem that image changes such as rotation and blur affect the registration effect. Literature [13] proposed a feature point matching algorithm combined with PSO, which effectively reduced the matching speed. From the enlarged image details, it can be seen that in the first image, the methods of [12] and [13] both show obvious stitching deviations in the white house area. In the third image, the method of document [12] appears to have blurred images. Compared with the other two algorithms, this paper uses STN-DenseNet model for feature extraction, which effectively improves the robustness of feature points, and uses improved GMS algorithm for feature matching, which reduces the matching speed and improves the matching accuracy. From the stitching results of the method in Figure 3 (c), there is no obvious geometric misalignment in the stitched images, the edges of the overlapping regions have good transitions, the brightness changes uniformly, and the overall imaging is natural.

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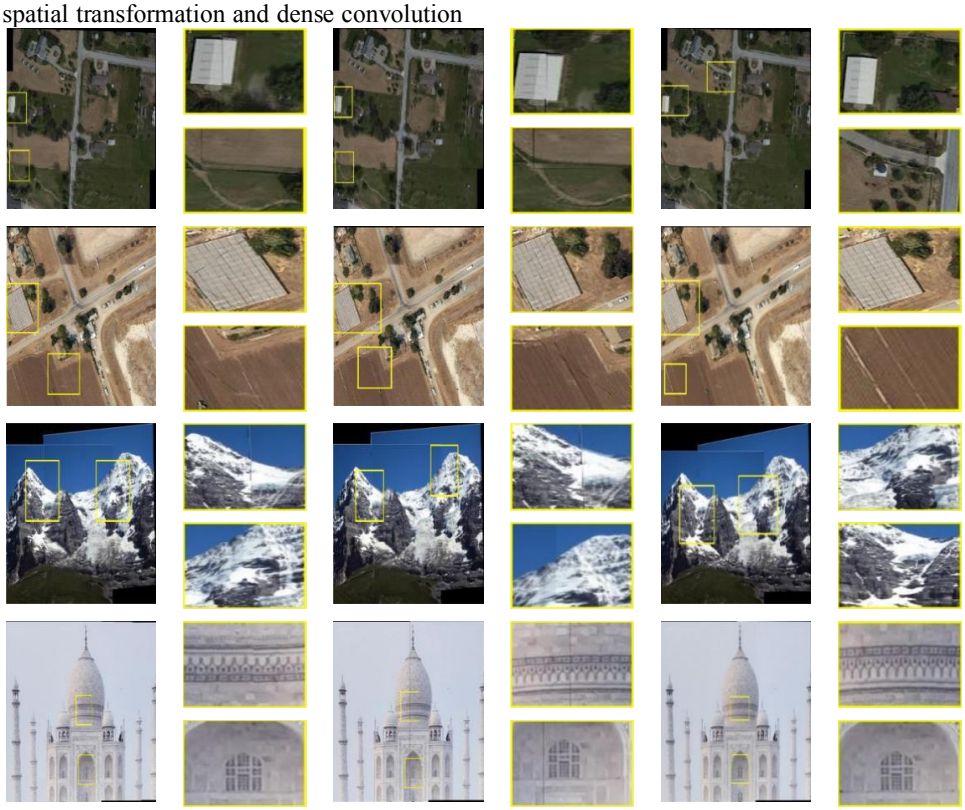
Table 2. Root mean square error （pixels）

Method	First group	Second Group	The third group	Fourth group
Literature [12]	0.56	0.49	0.58	0.61
Literature [13]	0.43	0.38	0.41	0.49
OURS	0.31	0.29	0.33	0.45

As shown in Table 2, the root mean square error of all algorithms is within a pixel range, while the root mean square error of this paper is smaller than other algorithms, and the registration accuracy is the highest. Compared with the literature [12] and literature [13] algorithms, the root mean square error is reduced by an average of 38.6% and 19.8% respectively, and the registration accuracy is significantly improved.

IV. CONCLUSION

This paper proposes a remote sensing image registration method based on spatial transformation and dense convolution fusion. In the feature extraction phase, feature extraction is performed through the densely connected network fused with a spatial transformation network, which effectively improves the robustness of feature points. In the feature point matching stage, an improved GMS algorithm is used for feature matching, and a method based on the homography matrix is used to eliminate mismatch points, which effectively improves the registration accuracy. Using the remote sensing image registration method proposed in this paper for image stitching, the synthesized panoramic image has no obvious geometric misalignment and blurring problems, and the edges of the overlapping areas have good transitions. Research and experiments show that the method in this paper has good robustness and registration effect on remote sensing images, and has certain practical research and application value. Compared with the other two methods, the root mean square error of the method in this paper is reduced by 38.6% and 19.8%. Registration accuracy has been improved significantly.



(a) Literature [12] (b) Literature [13] (c)OURS

Fig.2 Remote Sensing images stitching for comparison

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